

Solutions for Final Practice Exam

December 2025

Problem 1

- **Answer:** No.
- **Conditions to be a valid vector space:**
 1. It contains the zero vector $(0, 0)$.
 2. It is closed under vector addition.
 3. It is closed under scalar multiplication.

For the set to be a valid vector space, three conditions must be satisfied. On the other hand, if at least one of these conditions is violated, the set cannot be a valid vector space.

- **Zero vector**

The zero vector is $(0, 0)$. Check if it belongs to the set. For $a = 0$, we need $b = 2(0) + 5 = 5$. So the vector is $(0, 5)$, not $(0, 0)$. Thus, the zero vector is not in the set.

- **Closure under addition**

Take two vectors in the set: $(a, 2a + 5)$ and $(c, 2c + 5)$. Their sum is $(a + c, 2a + 5 + 2c + 5) = (a + c, 2(a + c) + 10)$. However, the condition for membership is $b = 2(a + c) + 5$. Here we get $b = 2(a + c) + 10$. So the sum is not in the set.

- **Closure under scalar multiplication**

Take $(a, 2a + 5)$ and multiply by scalar $k \in \mathbb{R}$:

$$k(a, 2a + 5) = (ka, 2ka + 5k)$$

For membership, we need $b = 2(ka) + 5$. However, we have $b = 2ka + 5k$. These are equal only if $k = 1$. So, closure fails in general.

- **Conclusion:**

The set $\{(a, b) \in \mathbb{R}^2 : b = 2a + 5\}$ is not a valid vector space, because it does not contain the zero vector, and is not closed under addition or scalar multiplication.

Problem 2

- **Answer:** Linearly independent.

- **Condition for the vectors to be linearly independent:**

The unique solution for the following linear system is $[0, 0, 0, 0]^\top$:

$$\begin{bmatrix} 2 & 2 & 3 & 1 \\ 1 & 2 & -1 & 1 \\ 1 & 1 & 6 & 1 \\ 5 & 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- **RREF of the augmented matrix:**

$$\begin{aligned} \left[\begin{array}{cccc|c} 2 & 2 & 3 & 1 & 0 \\ 1 & 2 & -1 & 1 & 0 \\ 1 & 1 & 6 & 1 & 0 \\ 5 & 1 & 1 & -1 & 0 \end{array} \right] &\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 2 & 2 & 3 & 1 & 0 \\ 1 & 1 & 6 & 1 & 0 \\ 5 & 1 & 1 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 \\ 1 & 1 & 6 & 1 & 0 \\ 5 & 1 & 1 & -1 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 5 & 1 & 1 & -1 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & -9 & 6 & -6 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & -2 & 5 & -1 & 0 \\ 0 & -9 & 6 & -6 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 13 & 1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & -2 & 5 & -1 & 0 \\ 0 & -9 & 6 & -6 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 0 & 13 & 1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & 0 & -9 & -1 & 0 \\ 0 & -9 & 6 & -6 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 13 & 1 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & 0 & -9 & -1 & 0 \\ 0 & 0 & -57 & -6 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & -4/9 & 0 \\ 0 & -1 & 7 & 0 & 0 \\ 0 & 0 & -9 & -1 & 0 \\ 0 & 0 & -57 & -6 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & -4/9 & 0 \\ 0 & -1 & 0 & -7/9 & 0 \\ 0 & 0 & -9 & -1 & 0 \\ 0 & 0 & -57 & -6 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & -4/9 & 0 \\ 0 & -1 & 0 & -7/9 & 0 \\ 0 & 0 & 1 & 1/9 & 0 \\ 0 & 0 & -57 & -6 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & -4/9 & 0 \\ 0 & -1 & 0 & -7/9 & 0 \\ 0 & 0 & 1 & 1/9 & 0 \\ 0 & 0 & 0 & 1/3 & 0 \end{array} \right] \\ &\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right] \end{aligned}$$

Therefore, $[0, 0, 0, 0]^\top$ is the unique solution.

Problem 3

- **Answer:** No.
- For three vectors to form a set of linearly independent vectors,

$$[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} v_{11} & v_{21} & v_{31} \\ v_{12} & v_{22} & v_{32} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

where the only possible solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$. For this linear system to have this unique solution, the number of pivots in the RREF of the augmented matrix, $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{0}]$, should be equal to not only the number of pivots in the RREF of the coefficient matrix, $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$, but also the number of unknowns (3). However, the number of pivots can be 2 at most for the RREFs of $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{0}]$ and $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$, which is less than the number of unknowns. Therefore, it is impossible for three vectors to be linearly independent in $\mathbb{R}^2$.

Problem 4

- **Answer:** Yes.
- If 4 vectors are independent, then none of them can be expressed as a linear combination of the others. Consider any subset of 3 of these 4 vectors:
 - If those 3 were dependent, then one of them could be expressed as a linear combination of the other two.
 - But that would mean the full set of 4 vectors is also dependent (because the dependent relation among the 3 would carry over when the 4th is added).
 - This contradicts the assumption that the original 4 are independent.

Therefore, every subset of 3 vectors must also be linearly independent.

Problem 5

- $\det(E)$
Since some or all of the columns are linearly dependent, $\det(E) = 0$.

- E^{-1}

If the determinant of a matrix is 0, the inverse matrix cannot exist.

- $\text{rank}(E)$

The rank of a square matrix is the number of linearly independent columns. Hence,

1. From the condition given by the problem, there can be at most $n - 1$ linearly independent columns (n : Number of columns):

$$\text{rank}(E) = n - 1$$

2. If all the columns are linearly dependent:

$$\text{rank}(E) = 1$$

Therefore, the rank of E is an integer in $[1, n - 1]$.

Problem 6

$$\int_0^{10} r(t)dt = \int_0^{10} 50 \exp(-0.1t)dt = [-500 \exp(-0.1t)]_0^{10} = 500 - \frac{500}{e}$$

Problem 7

P and Q are equivalent statements. Therefore, each statement is necessary and sufficient condition for the rest one.

Problem 8

For $\mathbf{u}$ and $\mathbf{v}$ to be orthogonal to each other, $\mathbf{u} \cdot \mathbf{v} = 0$. Hence,

$$\begin{aligned} 1 \times 2 + 2 \times (-1) + (-1) \times k &= 0 \\ \therefore k &= 0 \end{aligned}$$

Problem 9

- **Steepest decrease:** Move opposite the gradient, i.e.,

$$\begin{aligned} -\nabla J(\theta_1, \theta_2) &= -(2\theta_1 + 4\theta_2, 4\theta_1 + 10\theta_2)^\top \\ -\nabla J(1, -1) &= (2, 6)^\top \end{aligned}$$

- **Normalized direction:**

$$-\frac{1}{\sqrt{2^2 + 6^2}} \nabla J(1, -1) = \frac{1}{\sqrt{10}} (1, 3)^\top$$

Problem 10

- $E[3X - 2Y]$

$$E[3X - 2Y] = 3E[X] - 2E[Y] = 3 \times 1 - 2 \times 4 = -5$$

- $\text{Var}(3X - 2Y)$

Since X and Y are independent of each other,

$$\begin{aligned} \text{Var}(3X - 2Y) &= \text{Var}(3X) + \text{Var}(-2Y) \\ &= 3^2 \times \text{Var}(X) + (-2)^2 \text{Var}(Y) \\ &= 9 \times 2 + 4 \times 3 \\ &= 30 \end{aligned}$$

Problem 11

- **Answer:** 100
- **Two different approaches:**

1. **Count eligible cases:**

- Cases where the number of women is 1:

$$\binom{6}{2} \times \binom{4}{1} = 60$$

- Cases where the number of women is 2:

$$\binom{6}{1} \times \binom{4}{2} = 36$$

- Cases where the number of women is 3:

$$\binom{6}{0} \times \binom{4}{3} = 4$$

2. **Simpler approach:** Subtract the cases of forming a team with three men from the all possible cases

$$\binom{10}{3} - \binom{6}{3} = 120 - 20$$

Problem 12

- **Base case:**

$$2^{1+2} + 3^{2 \times 1 + 1} = 8 + 27 = 35 = 7 \times 5$$

- Let us assume that $P(n = k)$ is valid:

$$2^{k+2} + 3^{2 \times k + 1} = 7m, \quad m \in \mathbb{Z}$$

- Now, we show that $P(n = K + 1)$ is true given $P(n = k)$ is valid (k is a positive integer):

$$\begin{aligned} 2^{(k+1)+2} + 3^{2 \times (k+1) + 1} &= 2^{k+3} + 3^{2k+3} \\ &= 2 \times 2^{k+2} + 9 \times 3^{2k+1} \\ &= 2 \times (2^{k+2} + 3^{2k+1}) + 7 \times 3^{2k+1} \end{aligned}$$

From the assumption above,

$$\begin{aligned} &= 2 \times 7m + 7 \times 3^{2k+1} \\ &= 7 \times (2m + 3^{2k+1}) \end{aligned}$$

Since $2m + 3^{2k+1} \in \mathbb{Z}$, $P(n = k + 1)$ is true given $P(n = k)$ is valid.

Therefore, the statement is true for all positive integers n by the base case and the recursive relationship between $P(n = k)$ and $P(n = k + 1)$.

Problem 13

- Let A be the events where the sum of two numbers is even. Then, the number of possible cases for A is 18 because for the sums to be even, two numbers should be either both odd numbers ($3 \times 3 = 9$) or both even numbers ($3 \times 3 = 9$).
- Let B be the events where the sum of two numbers is 8. Out of 18 cases of A , the sum of two numbers is 8 when two numbers are one of the following 5 pairs:

$$(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$$

Therefore,

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{5/36}{18/36} = \frac{5}{18}$$

Problem 14

- $P(\bar{X}_{1000} > 47)$

$$P(\bar{X}_{1000} > 47) = P\left(Z = \frac{\bar{X}_{1000} - \mu}{\sigma/\sqrt{n}} > \frac{47 - 45}{45/\sqrt{1000}} = \frac{4\sqrt{10}}{9} \approx 1.41\right) \approx 0.07927$$

- $P(\bar{X}_{1000} < k) = 0.1$

$$P(\bar{X}_{1000} < k) = P\left(Z = \frac{\bar{X}_{1000} - \mu}{\sigma/\sqrt{n}} < \frac{k - \mu}{\sigma/\sqrt{n}} = \frac{k - 45}{45/\sqrt{1000}}\right) = 0.1$$

Hence,

$$\begin{aligned}\frac{k - 45}{45/\sqrt{1000}} &\approx -1.28 \\ k - 45 &\approx -1.28 \times \frac{9}{2\sqrt{10}} \\ &\approx -1.82 \\ \therefore k &\approx 43.18\end{aligned}$$

Problem 15

- **Sample mean:**

$$\frac{8.6 + 9.4 + 7.9 + 6.8 + 8.3 + 7.3 + 9.2 + 9.6 + 8.7 + 11.4 + 10.3 + 5.4 + 8.1 + 5.5 + 6.9}{15} \approx 8.227$$

- **Sample variance and sample standard deviation:**

$$s^2 = \frac{1}{15 - 1} \sum_{i=1}^{15} (x_i - \bar{x})^2 \approx 2.796, \quad s \approx 1.672$$

- **95% confidence interval:**

Since we do not know the population variance (i.e., population standard deviation), we need to use Student's t -distribution to construct the confidence interval:

$$\begin{aligned}\bar{x} \pm \text{margin} &= \bar{x} \pm t_{0.025, n-1} \times \frac{s}{\sqrt{n}} \\ &\approx 8.227 \pm 2.145 \times \frac{1.672}{\sqrt{15}}\end{aligned}$$